Shannon Channel Capacity Bit Rate

$$\text{Shannon channel capacity } C = B \cdot \log_2(1 + SNR) \quad \text{(in bits/second)}$$

Consider the old landline telephone communication system with a bandwidth *B* = (3,400 – 300) Hz = 3,100 Hz. The telephone system is generally assumed that to require a *SNR* of at least 3162 to operate without unacceptable bit errors. Calculate the maximum channel transmission capacity *C* as predicted by the Shannon equation.

Solution:
We know that the bandwidth *B* = 3,100 Hz, and a *SNR* is numerically equal to 3,162.

$C = 3{,}100 \times \log_2(1 + 3{,}162) = 3{,}100 \times \log_2(3{,}163)$
$= 3{,}100 \times 11.6271 = 36{,}044$ bits/sec = 36.044 kbps

Nyquist Bit Rate

$$\text{Nyquist channel capacity } C = 2B \times \log_2(L) \quad \text{(bits/second)}$$

where L is the number of signal levels per symbol

Calculate the Nyquist data rate given the same bandwidth B = 3,100 Hz and assume 8 signal levels per symbol.

Solution: Substituting these values into the Nyquist equation gives

Nyquist channel capacity C= 2* 3,100* log2 (8)
=18600 bps

Example:

Consider an extremely noisy channel which in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel, calculate the channel capacity C.

Solution:

Capacity $= \text{Bandwidth} \times \log_2(1 + \text{SNR})$ bits/sec
$= \text{Bandwidth} \times \log_2(1 + 0)$ bits/sec
$= 0$ bits/sec

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example:

Calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz. The signal-to-noise ratio is usually 3162.

Solution:

The channel capacity can be calculated as

Capacity $= \text{bandwidth (B)} \times \log_2(1 + \text{SNR})$ bits/sec
$= 3000 \times \log_2(1 + 3162)$ bits/sec
$= 34{,}860$ bits/sec or bps

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. Calculate the maximum bit rate.

Solution:

Bit rate (R) = $2\times3000\times\log_2 2 = 6000$ bps

Example:

Suppose we need to send 265 kbps over a noiseless channel with a bandwidth of 20kHz. How many signal levels do we need?

Solution:

$265000 = 2\times20000\times\log_2 L$

or, $\log_2 L = 6.625$

or, $L = 98.7$ levels

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example:

The signal-to noise ratio is often given in decibels =3.6 and the channel bandwidth is 2MHz. Calculate the theoretical channel capacity:

Solution:

SNR in dB = $10 \log_{10}$SNR

or, $3.6 = 10 \log_{10}$SNR

or, SNR = $10^{3.6/10}$

or, SNR = 2.29

$C = B\times\log 2(1 + SNR)$

$= 2\times10^6\times\log_2(1+2.29) = 3.436$ Mbps

1. **Find the capacity of a channel for a signal with bandwidth of 3.1 KHz and signal to noise ratio of 0 dB and comment on it.**

Solution:

$BW = 3.1$ KHz, $(SNR)_{dB} = 0$ dB

We know,

$(SNR)_{dB} = 10 \log_{10} SNR$

or, $0 = 10 \log_{10} SNR$

$\therefore \quad SNR = 1$

Now,

$C = BW \times \log_2 (1 + SNR)$

$= 3.1 \times \log_2 (1 + 1)$

$= 3.1$ Kbps

2. **Calculate the channel capacity having bandwidth and SNR of 6 Hz and 6 dB respectively.**

Solution:

Bandwidth (B) = 6Hz

SNR = 6 dB

or, $10\log\left(\frac{S}{N}\right) = 6$

$\therefore \frac{S}{N} = 10^{0.6} = 3.981$

6=10log10(SNR)

$SNR = 10^{\frac{6}{10}}$

SNR=3.981

The channel capacity is given by

$C = B \log_2 (1+SNR)$

$= 6 \times \log_2(1 + 3.981)$

$= 13.92$ bits/sec

The spectrum of a channel is in between 3 MHz and 4 MHz, and SNR = 24 dB. Calculate the bandwidth, the maximum channel capacity using Shannon's formula. Then based on Nyquist's formula, find the number of signal levels required to reach the maximum channel capacity.

Solution:

Bandwidth (B) = 4 MHz − 3 MHz = 1 MHz = 10^6 Hz

We have,

$(SNR)_{dB} = 10 \log_{10}(SNR)$

or, $24 = 10 \log_{10}(SNR)$

or, SNR = 251.18

Now,

$C = B \log_2(1 + SNR)$

$= 10^6 \times \log_2(1 + 251.18) \approx 10^6 \times 8 = 8$ Mbps

From Nyquist formula,

$C = 2B \log_2 L$

or, $8 \times 10^6 = 2 \times 10^6 \log_2 L$

or, $4 = \log_2 L$

$\therefore L = 16$

On a Tx. channel, 600 character messages using ASCII 7 bit code is used. For sync. data ~~frame~~ stream, there are two SYN character & a single error detection character is added. In async. data Tx, there are one ~~bit~~ start-bit & one stop-bit & single error detection character is added, Calculate the efficiency of transmission in both Tx. mode.

Soln:

For Sync. Tn. mode :

Original information character = 600 character

Total no. of characters transmitted
= 600 + 2 SYN character + 1 error detection character
= 600 + 2 + 1
= 603 character.

Then,
total no. of bits transmitted = 603 × 7
= 4221 bits

& actual data bits = 600 × 7 = 4200 bits

Thus,
efficiency $\eta = \frac{4200}{4221} \times 100\%$

= 99.5 %.

For Async. Tn mode :

Original info. character = 600 character

Total no. of characters transmitted = 600 + 1 Error detection characters
= 601 characters

Here,
each character is represented by :
7 ASCII bits + 1 start bit + 1 stop bit
= 9 bits

Then,
Total no. of bits transmitted = 601 × 9
= 5409 bits

Actual data bits = 600 × 7 = 4200 bits

efficiency $(\eta) = \frac{4200}{5409} \times 100\% = 77.65\%$

Example 1. **A company must transmit Parallel 12 M bytes file in 25 or fewer minutes. What is the minimum speed (bits per second) necessary to send the entire file? Assume that each byte is 8 bits and error rates and overhead are not significant factor. Calculate the minimum speed.**

Solution:

Data to be send = 12 Mbyte = 12×10^6 bytes

= $12 \times 10^6 \times 8$ bits

Given time = 25 minutes = 25 x 60 seconds

= 1500 sec

Now it is required to send 96×10^6 bits in 1500 seconds resulting in data rate of

(96×10^6)bps/ 1500 = **64 Kbps**

Q. A company has provide bandwidth of 64 kbps. Calculate how large file you can download in 10min.

Solution :

File download in 1 min = 64

File download in 10 min = $64 \times 10^3 \times 10 \times 60$

= 38.4 MB

The file in Byte from = $\frac{38.4}{8}$ =4.8MB

So, 4.8 MB file can be download in 10 min.

Q. 2 A channel is transmitting at an speed of 1 H bits per second in a serial synchronous mode. Calculate how many extra bits will be receive if a clock has become faster by .001%.

Solution : 1 mbps $=10^6$ bits

If a clock has become faster by .001% then

$$= .001 \times \frac{10^6}{100} = 10\text{bits}$$

Then, we will receive 10 more bits.

Problem 2. Compare the efficiency of sending a 256-byte message synchronously and asynchronously. The synchronous transmission need an 8-byte preamble for clock recovery.

Solution. In asynchronous transmission 10-bit are sent in place of 7 bits transmission. There is one start, eight data bits (7 bits + 1 parity bit), and one stop bits. Therefore, the efficiency of asynchronous transmission is given as -

Efficiency = Data bits/Total bits x 100% = (7/10) x 100% = 70%

The total number of bytes in the synchronous message is 256 data plus 8 preamble or 264 bytes. Thus, the efficiency of the synchronous message becomes -

Efficiency = (256/264) x 100% = 96.97%

Thus, the **synchronous transmission is more efficient.**

Problem 3. Compare the efficiency of sending the synchronous and asynchronous E character.

Solution. Here, we know that the synchronous data uses just the seven bits required for the E character's code while the asynchronous stream needs 10 bits in place of 7 bits transmission. There is one start, eight data bits (7 bits + 1 parity bit), and one stop bits.

Since all of the synchronous bits are data bits, the efficiency of sending the synchronous E is 8/8 x 100% = 100% (Note : one parity bit is also added in 7 bit tranmission)

On the other hand, the asynchronous has a much less efficiency = $\frac{7 \times 100}{10}$ % = 70%

Thus, **synchronous transmission is more efficient.**